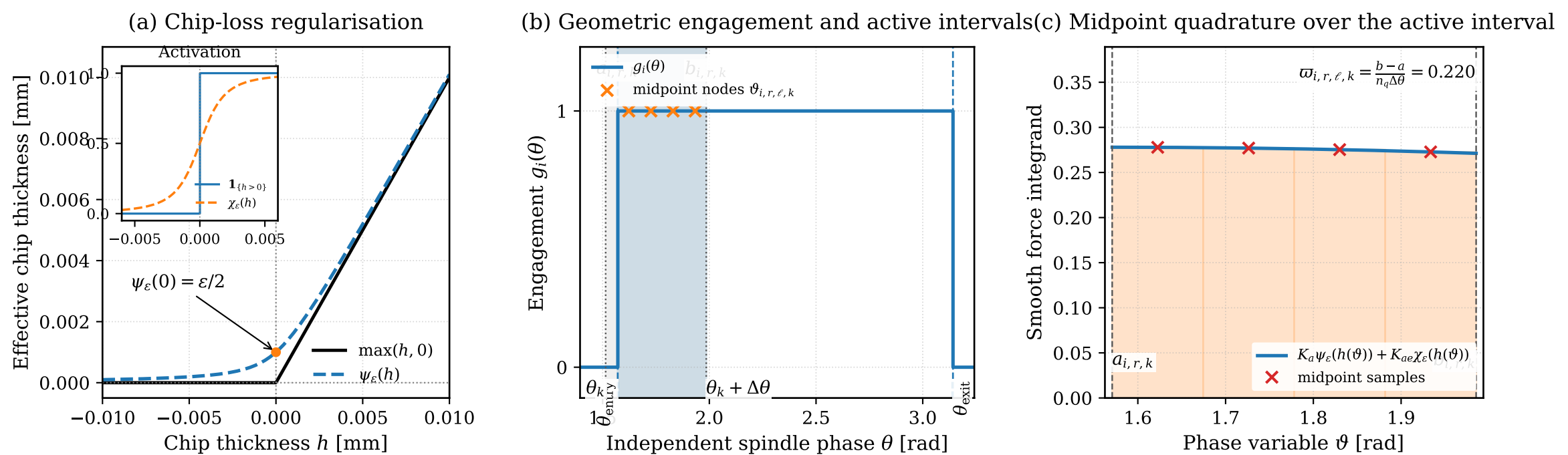
(a) Chip-loss regularisation
Activation
$\mathbf{1}_{\{h>0\}}$
$\chi_\varepsilon(h)$
$\psi_\varepsilon(0) = \varepsilon/2$
max(h, 0)
$\psi_\varepsilon(h)$
Effective chip thickness [mm]
Chip thickness h [mm]
(b) Geometric engagement and active intervals
$g_i(\theta)$
midpoint nodes $\vartheta_{i,r,\ell,k}$
Engagement $g_i(\theta)$
θ_k
$\theta_k + \Delta\theta$
θ_{entry}
θ_{exit}
Independent spindle phase θ [rad]
(c) Midpoint quadrature over the active interval
$\varpi_{i,r,\ell,k} = \frac{b-a}{n_q \Delta\theta} = 0.220$
$a_{i,r,k}$
$K_a\psi_\varepsilon(h(\vartheta)) + K_{ae}\chi_\varepsilon(h(\vartheta))$
midpoint samples
Smooth force integrand
Phase variable ϑ [rad]